

Predicting Sleep Apnea using Machine Learning: A Comparative Study based on User Survey Data

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ABSTRACT: Millions of people worldwide suffer from a dangerous sleep disorder called sleep apnea. It is characterized by frequent pauses in breathing as you sleep. Conventional diagnostic techniques, like overnight polysomnography, are expensive, time-consuming, and necessitate medical supervision. In order to provide a practical and affordable option for early diagnosis, this project intends to create a machine- based predictive method for identifying sleep apnea using user survey data. A Google Form is used to gather information on variables such as BMI, sleep duration, snoring frequency, and daytime fatigue. The risk of sleep apnea is predicted using a variety of machine learning techniques, such as SVM, DTC, Random Forest ANN, LR, and RF. The model with the highest forecast accuracy is selected for deployment. By providing a pre-screening tool for AI-powered sleep apnea detection that anyone can use without requiring clinical intervention, this method improves accessibility and affordability.

Keywords: web application, health screening, machine learning, prediction model, sleep apnea, and user survey.

I-INTRODUCTION

Sleep apnea is a common but underdiagnosed sleep disorder that can lead to serious health issues like heart disease, hypertension, and daytime fatigue. Traditional diagnostic methods like polysomnography are out of reach for many people, especially in rural areas. This study uses ML models built on user-reported survey data to suggest an affordable, non-invasive method of early detection. Age, BMI, frequency of snoring, and medical problems like diabetes and hypertension are among the demographic, physiological, and lifestyle characteristics that are included in the study. Accuracy, precision, recall, and F1- score were used to assess a number of machine learning methods, including LR, DT, RF, SVM, KNN, Gradient Boosting, and a Stacking Classifier. Techniques such as RFECV for feature selection and SMOTE Tomek for data balancing were used to enhance model performance. Finding the most dependable

model for actual application in early screening is the goal of the project. The method is a useful first step to increase awareness and assist medical experts, but it cannot take the place of a medical diagnosis.

II-LITERATURE REVIEW

This study demonstrates how polysomnography data may be used to support the diagnosis for obstructive sleep apnea utilizing deep learning and machine learning models. Explainable artificial intelligence seeks to make these models easier to understand, despite the fact that they are frequently seen as "black boxes." In this work, machine learning techniques are used to assess respiratory signals and associated features. Model predictions are explained using the Local Interpretable Model-Agnostic Explanations (LIME) approach. The findings demonstrate how various signal characteristics affect apnea detection and sleep quality [1].

This paper shows a machine-learning method that uses electronic health data to predict obstructive sleep apnea. The suggested model combines the BiLSTM network for the classification with XGBoost for identifying the most significant risk factors. With just six essential features, the model achieves 97% accuracy utilizing data from the Swedish National Study on Aging along with Care. Performance is contrasted using LSTM and other cutting-edge techniques. The findings demonstrate increased accuracy in predicting and improved detection of sleep apnea risk variables [2].

This paper describes that using polysomnographic data, four machine learning algorithms SVR, kNN, Random Forest, and Gradient Boosting are used to predict the degree of symptoms of obstructive sleep apnea. Recursive feature elimination and feature engineering were used to lessen overfitting and increase model generalization. Robustness was ensured by a 5-fold cross-validation technique, and Random Forest achieved an extremely high out-of-bag score of 0.97. Along AUCs of 1.00 and 0.99, respectively, Random Forest and SVR performed the best. The findings show how machine learning models can help with trustworthy clinical judgment in OSA evaluation [3].

This study shows a machine learning-based method that uses sleep related and lifestyle characteristics to predict sleep apnea. MATLAB's Regression Learner was used to assess variables including daily step count, physical activity, level of stress and sleep quality. Key patterns impacting sleep apnea were found through data preprocessing and visualization. The findings demonstrate that

machine learning is capable of accurately capturing how lifestyle choices affect sleep quality. The paper emphasizes the possibility of early identification and future integration into real time embedded systems and deep learning[4].

This study looks into machine learning techniques as affordable substitutes for polysomnography in the diagnosis of sleep apnea and insomnia. For the categorization of sleep disorders, Random Forest, bagging along with SVM and also Logistic Regression were assessed. 5-fold cross-validation confirmed that bagging had a maximum accuracy of 92.4%. Model performance was further enhanced via hyperparameter adjustment. Sleep duration, quality of sleep and BMI were found to be important prognostic factors[5].

This paper proposed a deep learning-based approach that uses single lead ECG readings to automatically diagnose obstructive sleep apnea. A deep neural network tuned by Bayesian optimization is coupled with wavelet transform with heart rate variability characteristics. The ECG5000 dataset is used to train and evaluate the model. It outperforms traditional approaches with an accuracy of 93.51%. The findings demonstrate how ECG based deep learning can be utilized by wearable healthcare systems to identify OSA in real time and at a reasonable cost [6].

This paper shows a new machine learning algorithm that uses electronic health data to detect obstructive sleep apnea. The method integrates an BiLSTM network in classification with XGBoost for identifying the most significant risk factors. With just six essential features, the model achieves 97% accuracy using data gathered by the Swedish National Study on Aging with Care. It performs better than other cutting edge models and traditional LSTM. The findings show better risk factor identification and diagnosis of sleep apnea [7].

In this study, deep learning techniques for anticipating obstructive sleep apnea episodes utilizing unprocessed respiratory signals are presented. To forecast apnea and hypopnea occurrences thirty seconds in advance, four models that utilize CNN and LSTM architectures have been set forth. CNN-LSTM variations performed the best, with all models achieving above 81% accuracy when tested using data from 1,507 participants. Low sample rates did not affect the models' efficacy, allowing for their adoption in inexpensive at-home devices. These findings show that proactive OSA prediction along with airflow control during sleep are feasible [8].

This review paper examines methods for discovering and diagnosing sleep apnea using deep

learning and machine learning. It examines research employing ECG, EEG and PPG signals, emphasizing various algorithms, feature extraction techniques, including multi-signal integration for increased precision. The review highlights significant developments, assesses present issues, and suggests future paths to improve robustness, interpretability with real time sleep apnea detection [9].

This study proposed a machine learning-based multi-parameter oximetry model to detect obstructive sleep apnea (OSA). ODI, MinSpO₂, ST90 along with hypoxia load are among the eight SpO₂-derived metrics that were integrated into several models and assessed using six machine learning techniques. With an accuracy of 94.5% and an AUC of 0.9885, the CatBoost-based ST90-ODI-MinSpO₂-HB model performed the best. ODI, HB, and MinSpO₂ were identified as important predictors by SHAP analysis. The model's promise for individualized, precise and easily accessible OSA screening was demonstrated by subgroup analysis, which revealed better performance in female and older subjects [10].

This paper proposed a useful sleep monitoring method to identify obstructive sleep apnea (OSA). To increase detection accuracy, the technology combines machine learning with SpO₂, heartbeat, body position and snore events. To categorize sleep occurrences as apnea, hypopnea or normal, it computes the Apnea Hypopnea Index (AHI). Early diagnosis and prompt treatment are made possible by a web interface that makes results accessible to both patients and physicians [11].

This study uses anthropometric as well as questionnaire data for predicting obstructive sleep apnea (OSA) using the multiclass Mahalanobis Taguchi system (MMTS). MMTS outperformed techniques including logistic regression, artificial neural networks, SVM, C4.5 and also rough sets with an accuracy of 84.38%. The findings imply that MMTS can help physicians predict OSA, minimize the need for expensive and time consuming polysomnography tests and maximize the use of medical resources[12].

This study uses wearable-derived physiological data, such as age, BMI, oxygen saturation as well as heart rate to assess classifiers using machine learning for early sleep apnea identification. Using preprocessing to correct for imbalance, a dataset of real and synthetic samples was used to compare Logistic Regression, SVM, Random Forest and XGBoost. With 98.25% accuracy, ensemble techniques in particular, XGBoost performed the best. The study shows that machine learning (ML) can offer scalable, data driven diagnostic tools to

prompt and easily accessible sleep apnea detection [13].

This paper presented DreamGuardian an AI driven technology that uses non-invasive wearable sensors to detect and forecast sleep apnea early. Real-time sleep analysis is made possible by the system's monitoring of physiological signs like movement, variability in heart rate, and breathing patterns. Key characteristics, such as heartbeat, BMI, gender, age, ESS score, AHI and oxygen saturation, were chosen for study after preprocessing data from sleep clinics and repositories. Machine learning algorithms offer a scalable, easily accessible and precise diagnostic solution by predicting whether sleep apnea is present or not[14].

This study shows that using polysomnography to be the gold standard, machine learning techniques are used to screen individuals for obstructive sleep apnea (OSA). 63 papers including techniques like logistic regression, SVM, neural networks, decision trees and many others were included out of 5,479 articles. Most models required external validation in large cohorts and standardized OSA criteria, despite their great performance (logistic regression employing age, waist circumference, ESS score and oxygen saturation, for instance, reached an AUC of 0.98). The results underline the necessity of thorough validation while highlighting the promise of machine learning (ML) for OSA screening [15].

III. WORK FLOW

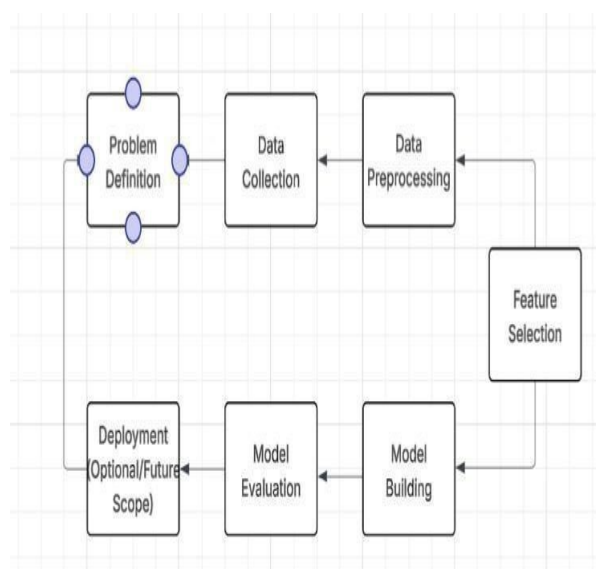


Fig.1. Workflow

Explanation for workflow:

Data Collection

A structured questionnaire was developed and circulated via Google Forms in both English and Tamil to gather real-world, user-reported data. The survey captured various health, lifestyle, and demographic attributes—such as age, gender, BMI, loud snoring, gasping for air, daytime sleepiness, medical history (e.g., diabetes, hypertension), sleep duration, alcohol consumption, and smoking habits.

Preprocessing of Data

Preprocessing entailed cleaning and transforming unprocessed survey information into a format that could be used by machine learning. This involved calculating BMI, standardizing height and weight units, and eliminating null data. Label and one-hot encoding were used to numerically encode responses in mixed languages.

Augmenting Data

Gaussian noise addition was used to generate synthetic data in order to increase the dataset's robustness and diversity. To account for minor variances, synthetic variants were developed from a base of actual survey data. The trained models became more generalizable across comparable but invisible patterns in real-world populations because to this augmentation, which also helped avoid overfitting.

Feature Selection

RFECV with Cross Validation, was used to select the most influential attributes affecting sleep apnea risk. This process ranked features such as BMI, snoring frequency, and gasping events based on their predictive power, thereby reducing dimensionality, improving model accuracy, and eliminating irrelevant or redundant information.

Split Test and Train

Eighty percent of the balanced and processed dataset was used for training, while twenty percent was used for testing. This made sure that performance of the model was assessed with data that had never previously observed, mimicking actual prediction situations. During the split, stratified sampling was employed to preserve class balance.

Make a Forecast

The chosen features were used to train a number

of machine learning models, such as Random Forest, SVM, KNN, Logistic regression combined with gradient boosting.

Visualization

Model performance and data interactions were interpreted using visual aids such confusion matrices, bar charts, and correlation heatmaps. Visualizations demonstrated class distribution, highlighted prominent risk factors (e.g., male gender, high BMI, short sleep duration), and effectively conveyed findings to both technical and non-technical audiences.

VI-MATHEMATICAL FORMULAE

Logistic Regression-uses the sigmoid function to translate a linear collection of features into a probability:

$$z = \beta_0 + \sum_{i=1}^n \beta_i x_i$$

$$(z) = \frac{1}{1 + e^{-z}}$$

Support Vector Machine (SVM):

The hyperplane $\mathbf{w}^T \mathbf{x} + \mathbf{b} = 0$ is sought. The margin between classes is maximized. The margin M is:

$$M = \frac{2}{\|\mathbf{w}\|}$$

The optimal $\mathbf{w}$ solves:

$$\min_{\mathbf{w}} \frac{1}{2} \|\mathbf{w}\|^2 \quad \text{s.t.} \quad y_i (\mathbf{w}^T \mathbf{x}_i + b) \geq 1 \quad \forall i$$

K-Nearest Neighbors (KNN)

Classifies a test point $\mathbf{u}$ by majority vote among its k nearest neighbors. Distance metric (Minkowski, p -norm):

$$d = (u, v) = \left(\sum_{i=0}^n |u_i - v_i|^p \right)^{1/p}$$

Common choices: $p = 2$ (Euclidean), $p = 1$ (Manhattan).

DecisionTree Splitting (Gini Index)

At each node, select feature that minimizes impurity. Gini impurity for node t :

$$C(t) = 1 - \sum p(c | t)^2$$

$$c=1$$

where $p(c | t)$ is the proportion of class c samples at t .

Gradient Boosting (Additive Model)

Builds M weak learners $h(\mathbf{x})$ in sequence to minimize loss

L . At iteration m :

$$(\mathbf{x}) = N_{m-1}(\mathbf{x}) + \gamma_m h_m(\mathbf{x})$$

where γ_m is the step size selected by line search in order to reduce $\sum_i L(y_i, F_{m-1}(\mathbf{x}_i) + \gamma h_m(\mathbf{x}_i))$.

Ensemble Stacking

Combines base model outputs $t^{(j)}$ into a meta-model:

$$t^{\wedge} = (t^{(\wedge 1)}, t^{(\wedge 2)}, \dots, t^{(\wedge J)})$$

where g is typically another classifier (e.g., Gradient Boosting) trained on the base predictions.

Evaluation Metrics

Let TP, TN, FP, and FN stand for true/false positives/negatives:

- accuracy $\frac{tp+tn}{tp+tn+fp+fn}$
- precision $\frac{tp}{tp+fp}$
- recall (Sensitivity) $\frac{tp}{tp+fn}$
- f1-Score $2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$

VII. Result AND Conclusion

With a maximum accuracy of 98% and nearly flawless precision and recall scores, the Stacking Classifier proved to be the most successful model, demonstrating its dependability for real-time sleep apnoea risk prediction. With respective accuracies of 70% and 69%, RF and LR were also among the individual models that performed well. SVM and gradient boosting models came in second and third, respectively, suggesting that the results were similar among classifiers. These results demonstrate how well ensemble learning approaches handle actual, survey-based health data. The main predictors of prediction accuracy were consistently identified as being BMI, snoring frequency, daytime drowsiness, and gasping for air.

Algorithm	Accuracy	0/1	Precision	Recall	F1 Score	Support
SVM	65%	0	0.78	0.85	0.81	124
		1	0.28	0.36	0.32	36
KNN	65%	0	0.74	0.86	0.80	124
		1	0.29	0.28	0.29	39
Gradient Boosting	67%	0	0.77	0.88	0.82	124
		1	0.31	0.31	0.31	36
Logistic Regression	69%	0	0.77	0.92	0.84	124
		1	0.34	0.31	0.32	36
Random Forest	70%	0	0.77	0.91	0.83	124
		1	0.44	0.31	0.36	36
Stacking Classifier(RF,KNN,GB)	98%	0	0.98	0.97	0.98	449
		1	0.98	0.99	0.98	449

Table1.Output Values

Diagrammatic representation of outputs

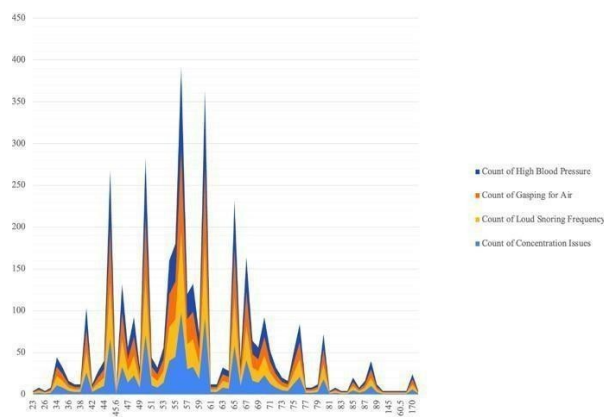


Fig 2. Age-wise Distribution of Sleep and Health Issues

The age-wise distribution of several health and sleep-related conditions, such as high blood pressure, difficulty breathing, frequent loud snoring, and difficulty concentrating, is depicted in this graphic. According to the findings, people between the ages of 50 and 60 had the highest incidence of all illnesses was

studied, indicating that middle age is a crucial time for the development or exacerbation of these health issues. Younger age groups (20–40 years) show comparatively fewer occurrences, however there are noticeable little surges. The prevalence of these problems progressively decreases as people age over 70, either as a result of fewer survey respondents or reduced symptom reporting in older populations. High blood pressure seems to be the most prevalent of the illnesses, followed by symptoms of sleep disorders such as gasping and snoring.

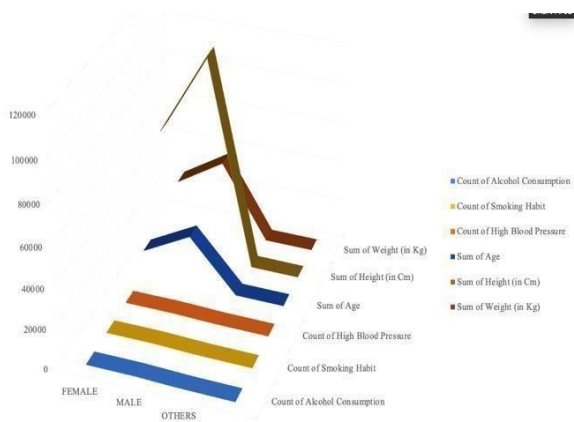


Fig 3. Comparison of Lifestyle Habits and Health Metrics

Three gender groups—Male, Female, and Others—are shown in the graphic along with a comparative summary of different lifestyle determinants and health measures. In addition to the sum of age, height, and weight, it also contains counts of high blood pressure, smoking, and alcohol use. In the majority of categories, male participants regularly report the highest values, especially in weight, height, and substance use, indicating a higher prevalence of these behaviors and the health concerns that go along with them. While the Others group has much lower totals, either as a result of a smaller sample size or lesser occurrence, female participants show lower but generally balanced values across all parameters. A significant increase in the weight total for men could be a sign of either a greater average body mass or more responders.

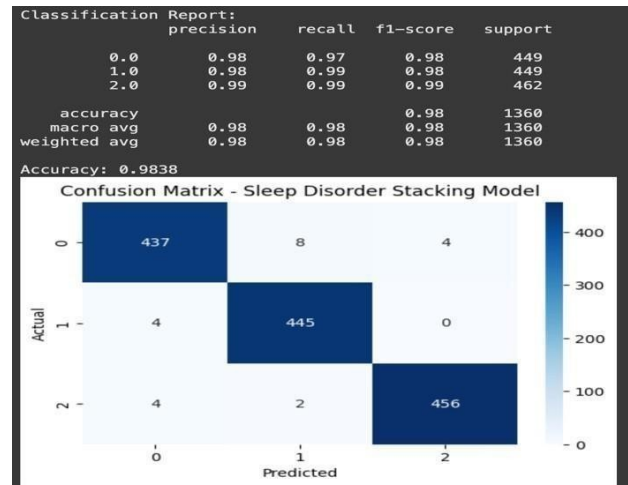


Figure 4. Confusion Matrix – Sleep Disorder Stacking Model

A thorough analysis of the Sleep Disorder Stacking Model's prediction ability across several sleep apnoea risk categories can be found in the Confusion Matrix. For each class, such as low, moderate, and high risk, it displays the proportion of accurate and inaccurate forecasts. With high true positive rates and few misclassifications, the matrix demonstrates the model's capacity to discriminate in between classes with accuracy. The model's overall 98% accuracy is supported by the diagonal dominance within the matrix, which shows good classification accuracy. This demonstrates how well the ensemble approach uses survey-based health and lifestyle data to reliably identify those who are at risk.

VI. FUTURE SCOPES

Expanding this study's influence and usefulness in the healthcare industry is its future scope. In order to provide real-time sleep apnea risk assessment for people without the need for clinical visits, the model can first be included into a mobile or web-based application. Furthermore, the model's generalizability will be improved by extending the dataset across various demographic groups and geographical locations. Additional factors like longitudinal sleep patterns and wearable sensor data (like heart rate and oxygen saturation) could increase the predicting accuracy. It is also possible to investigate more sophisticated deep learning models, such as CNNs or LSTM, to capture temporal dependencies in sleep behavior. Clinical relevance will be strengthened by working with medical experts to validate model predictions against medical diagnosis. Furthermore, including explainable AI (XAI) techniques can increase model decisions.

transparency and credibility, which makes them more appropriate for use in public health. All things considered, the system appears to be a promising low-cost, scalable method for early sleep apnea screening.

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